

WT Experiment No. 5

Name	Roll No
Palak Chanchlani	05
Sonam Chhabaidiya	09
Mahi Jodhani	21

Aim:

Simulation of Automatic Locking Mechanism Using RFID Module and LCD I2C in Tinkercad

Theory:

Tinkercad is an online simulation tool that allows users to design, simulate, and test electronic circuits. It provides a virtual environment where components like Arduino, sensors, LCDs, and more can be connected and programmed without physical hardware. In this experiment, we simulate an automatic locking system using an RFID module (RC522) and an LCD I2C. When an authorized RFID card is scanned, the system unlocks by moving the servo motor and displays "Access Granted" on the LCD. If an unauthorized card is scanned, it displays "Access Denied". This mechanism is widely used in secure entry systems, smart locks, and industrial automation.

Objectives of Simulation:

- To simulate an RFID-based locking system in Tinkercad.
- To display access status on an LCD I2C screen.
- To control a servo motor based on RFID authentication.
- To debug and test the circuit before real-world implementation.

Steps:

Step 1: Open Tinkercad and Create a New Circuit

1. Go to Tinkercad (<https://www.tinkercad.com/>).
2. Click on "Circuits" from the left panel.
3. Click "Create New Circuit".

Step 2: Add Components

Search and drag the following components onto the workspace:

1. Arduino Uno
2. RFID-RC522 Module
3. Servo Motor (SG90 or similar)
4. LCD 16x2 (I2C Module attached)
5. Jumper Wires

Step 3: Wire the Components

- ◆ Wiring the RFID Module (RC522) to Arduino

RFID Pin Connect To (Arduino)

VCC	3.3V
GND	GND
RST	Pin 9
SDA (SS)	Pin 10
SCK	Pin 13
MOSI	Pin 11
MISO	Pin 12

- ◆ Wiring the Servo Motor

Servo Pin Connect To (Arduino)

VCC (Red)	5V
GND (Black/Brown)	GND
Signal (Yellow/Orange)	Pin 6

- ◆ Wiring the LCD I2C

LCD I2C Pin Connect To (Arduino)

VCC	5V
GND	GND
SDA	A4
SCL	A5

Step 4: Write the Arduino Code

1. Click on "Code" and select "Text" mode.
2. Paste the following code into the Arduino editor:

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <SPI.h>

#include <MFRC522.h>

#include <Servo.h>

// LCD I2C address 0x27 (change if needed)
LiquidCrystal_I2C lcd(0x27, 16, 2);

#define SS_PIN 10

#define RST_PIN 9

#define SERVO_PIN 6

Servo lockServo;

MFRC522 rfid(SS_PIN, RST_PIN);

byte authorizedUID[] = {0xA1, 0xB2, 0xC3, 0xD4}; // Replace with your card's UID

void setup() {
    Serial.begin(9600);

    SPI.begin();

    rfid.PCD_Init();

    lcd.begin();

    lcd.backlight();

    lockServo.attach(SERVO_PIN);

    lockServo.write(0); // Locked position

    lcd.setCursor(0, 0);

    lcd.print("Scan RFID Card");

    Serial.println("Place your RFID card...");
}

void loop() {
    if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
        return;
    }
}
```

```

Serial.print("Card UID: ");

bool accessGranted = true;

for (byte i = 0; i < 4; i++) {
    Serial.print(rfid.uid.uidByte[i], HEX);
    Serial.print(" ");

    if (rfid.uid.uidByte[i] != authorizedUID[i]) {
        accessGranted = false;
    }
}

Serial.println();

lcd.clear();

if (accessGranted) {
    Serial.println("Access Granted");
    lcd.setCursor(0, 0);
    lcd.print("Access Granted");
    lockServo.write(90); // Unlock
    delay(3000);
    lockServo.write(0); // Lock again
} else {
    Serial.println("Access Denied");
    lcd.setCursor(0, 0);
    lcd.print("Access Denied");
    delay(2000);
}

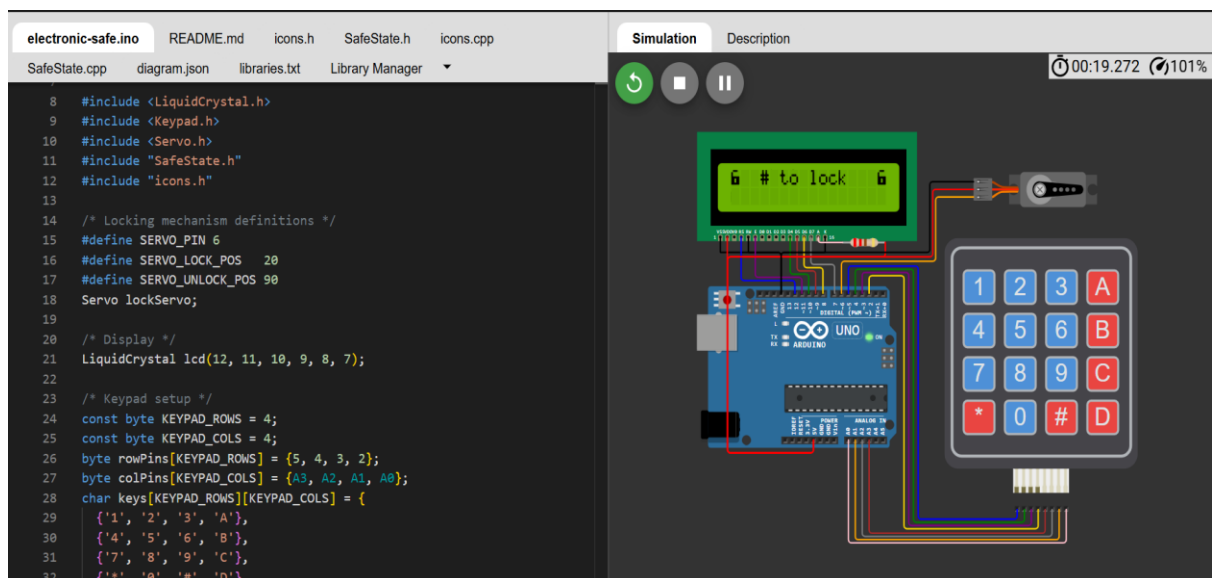
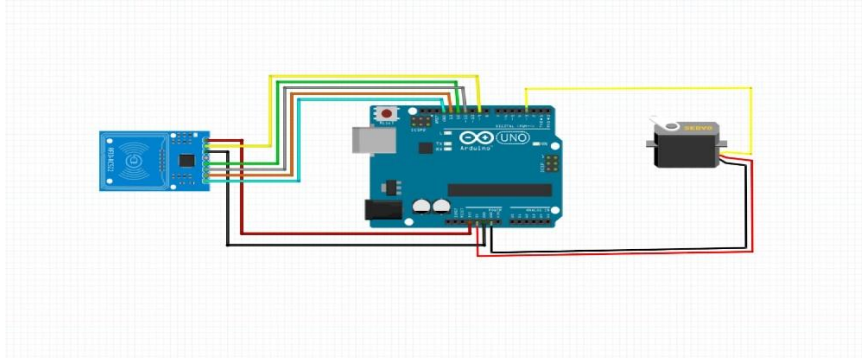
rfid.PICC_HaltA();
rfid.PCD_StopCrypto1();
}

```

Step 5: Start the Simulation

1. Click the "Start Simulation" button.
2. Simulate scanning an RFID card:
 - If the card UID matches → LCD displays "Access Granted", servo unlocks.

- If the card UID does not match → LCD displays "Access Denied", servo remains locked.
3. Open Serial Monitor (Click "Code" → "Monitor") to check UID values.
 4. If needed, replace authorizedUID[] in the code with your card's UID.



Conclusion:

The simulation successfully demonstrated an RFID-based automatic locking system using Arduino, an LCD I2C, and a servo motor in Tinkercad. The system correctly:

- Scanned RFID cards and compared their UID.
- Displayed authentication results on an LCD I2C.
- Controlled a servo motor for locking/unlocking.